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ORIGINAL INVESTIGATION

Night Games and Sleep: Physiological, Neuroendocrine, and Psychometric Mechanisms

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Context: Night games are a regular occurrence for team-sport athletes, yet sleep complaints following night competitions are common. The mechanisms responsible for reported sleep difficulty in athletes are not understood. ***Methods:*** An observational crossover design investigating a night netball game and a time-matched rest day in 12 netball athletes was conducted to ascertain differences in physiological (core temperature), psychometric (state and trait), and neuroendocrine (adrenaline, noradrenaline, and cortisol) responses. ***Results:*** Following the night game, athletes experienced reduced sleep durations, lower sleep efficiency, early awakenings, and poorer subjective sleep ratings compared with the rest day. No differences were found between core temperature, state psychometric measures, and cortisol at bedtime. Adrenaline and noradrenaline concentrations were elevated compared with the time-matched rest day prior to (26.92 [15.88] vs 12.90 [5.71] and 232.6 [148.1] vs 97.83 [36.43] nmol/L, respectively) and following the night game (18.67 [13.26] vs 11.92 [4.56] and 234.1 [137.2] vs 88.58 [54.08] nmol/L, respectively); however, the concentrations did not correlate to the sleep variables (duration, efficiency, and sleep-onset latency). A correlation ($r_s = -.611$) between sleep efficiency and hyperarousal (trait psychometric measure) was found. ***Conclusions:*** Athletes experienced poor sleep following a night game. Furthermore, results suggest that athletes who have a tendency toward a high trait arousal may be more susceptible to sleep complaints following a night game. These data expand knowledge and refute frequently hypothesized explanations for poor sleep following night competition. The results may also help support staff and coaches target strategies for individual athletes at a higher risk of sleep complaints.

Keywords: competition, netball, athletes, team sport, recovery

Despite the acknowledged importance of sleep for performance and recovery,[1] athletes commonly experience sleep loss following late competitions.[2–4] Specifically, team-sport athletes such as male footballers[4] and Australian rules footballers[5,6] have reported reduced sleep quantities of 181, 126, and 171 minutes, respectively, following night games. Given the regularity of night games for athletes, it is essential that the mechanisms responsible for disturbed sleep are understood, so targeted interventions can be implemented.

Sleep is a complex phenomenon, regulated by several regions of the brain through modulations of neurotransmitters and neuropeptides that control daily cycles of wakefulness and sleep.[7] The ascending arousal system in the hypothalamus and sleep-active neurons in the ventrolateral preoptic nucleus interact much like a "flip-flop switch," turning on and off periods of sleep and wakefulness.[8] Athletic competitions have the ability to stimulate neurotransmitters (adrenaline and noradrenaline) of the ascending arousal system, triggering the release of cortisol and potentially causing a disruption of the sleep–wake cycle.[9] Indeed, within patients with insomnia, high levels of cortisol prior to bedtime are consistent with disrupted sleep (increased nocturnal awakenings), confirming an interaction between sleep and the hypothalamic–pituitary–adrenal axis.[10]

Within sport, large increases in cortisol concentrations have been reported during competitions.[11] However, to date, no studies have explored neurochemical measures of the hypothalamic–pituitary–adrenal axis (cortisol, adrenaline, and noradrenaline) postexercise on subsequent sleep. Within civil servants[12] and those with insomnia,[10] increased urinary catecholamines and increased psychological arousal are associated with poor sleep quality. It is therefore plausible that the demands of athletic competition could stimulate alertness and vigilance into an athlete's sleep period, providing rationale for the reported decrease in sleep following late competitions.[13]

Given the documented deficiency in sleep (duration and quality) within team sport athletes following late competitions,[2–4] the lack of literature available to provide a mechanistic explanation indicates a need for research. Therefore, this study examined commonly hypothesized mechanisms thought to impact sleep (cortisol, adrenaline, noradrenaline, core temperature, and psychometric measures) following a night game and a time-matched control day in elite netballers.

Methods

Participants

Twelve Australian netball athletes (mean [SD]: age = 19.2 [0.9] y, height = 184.1 [6.9] cm, and body mass = 72.8 [5.2] kg) volunteered to participate in the study. All athletes were members of their relevant age-group national representative team. Over a 1-week period prior to the study, sleep diaries were completed that identified, on average, participants' required 8.5 (0.7) hours per night to feel rested, scored 8 (4) for the Epworth Sleepiness Scale, and rated their current sleep satisfaction as 8 (1) out of 10 on a visual analog scale. All participants were provided, in writing, with the risks and benefits of their participation, and informed consent was obtained from all individual participants included in the study.

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The Australian Institute of Sport Ethics Committee and Murdoch University Human Research Ethics Committee approved this study.

Design

An observational cross-over design was employed to ascertain differences in physiological and hormonal responses to an international night netball game (game starting at 6:00 PM)[4] and its consequent effect on sleep, compared with a time-matched rest day. The time-matched rest day was conducted 10 days following the night game. Participants were asked not to engage in any physical activity during the rest day. In addition, participants were familiarized with the study procedures and collection times before the collection period (Figure 1) and were encouraged to partake in normal nutritional protocols during the study. The sport of netball was selected for this study because it is common for games to occur late in a day and to be played over consecutive days within a tournament setting. A game of netball is played over a 60-minute duration and places high physical demands on players via repeated rapid accelerations and decelerations, explosive jumps, contact trauma, and muscle damage from eccentric loading. To partake in the study, athletes had to have played between 45 and 60 minutes of the night game. Sleep was monitored for 2 weeks throughout the study using Actical® actigraphy monitors (Philips Respironics, Bend, OR). No sleep education was provided prior to or during the sleep monitoring period. Each testing session (night game and rest day) consisted of a waking salivary cortisol measure followed 30 minutes later by an additional measure to calculate cortisol awakening response (the difference between the second and first morning sample). Salivary cortisol measures were later sampled 10 minutes prior to the game, 10 minutes postgame, 1 hour postgame, at bedtime, and upon awakening the next morning (Figure 1). In addition, commencing from waking, 24-hour urinary catecholamines (adrenaline and noradrenaline) were measured. Core temperature was recorded continuously in 1-minute intervals via a radio telemetric core temperature pill (HQ Inc, Palmetto, FL) from the cessation of the netball game (time matched on the control day) until awakening the following morning (ad libitum). All pills were calibrated and checked at 3 different temperatures against a certified mercury thermometer in a water bath prior to use. Four hours prior to the game, the core temperature pill was ingested to ensure that it was not sensitive to ingested fluid. No hydrotherapy was conducted following the netball game to avoid temperature interferences from immersion. A Presleep Arousal Scale (PSAS) was administered prior to bed to assess measures of state arousal, and the Karolinska Sleep Questionnaire was completed upon awakening to determine subjective sleep perceptions from the previous night's sleep.

During the rest day, data collection was time matched to the competition day, with the exclusion of the waking and prebedtime measures (self-selected to the body's natural rhythm) to account for any diurnal variations.

Sleep Analysis

Participants completed 2 weeks of sleep assessment via sleep diaries and wrist actigraphy monitors. Each actigraphy monitor contained a piezoelectric accelerometer, which sampled activity counts in 1-minute epochs. Epoch-for-epoch concordance rates of 81% to 90% with polysomnography (gold standard sleep measure) have been found for sleep/wake activity using Actical® wrist actigraphy monitors.[14] Participants wore the monitor at all times unless showering or competing. Participants recorded bedtime, wake time, and sleep quality ratings for each sleep period in a sleep diary the morning after the sleep period and selected sleep quality ratings from a 5-point scale of "very good," "good," "average," "poor," and "very poor." Diary information along with actigraphy data (event markers) was used to determine sleep/wake periods.[15] Individual nights of sleep were analyzed in Actiware® sleep software (version 5.61; Respironics Inc, Mini Mitter Philips/Respironics, OR) with the threshold sensitivity set to "medium."[16] Bedtime, wake time, sleep duration (amount of time spent asleep), sleep onset latency (amount of time between bedtime and sleep onset), number of wake bouts, and sleep efficiency (sleep duration expressed as a percentage of time in bed) were calculated following the rest and night game.

Neuroendocrine Assessment

Salivary Cortisol Assessment. Salivary cortisol samples were measured using Salimetrics oral swabs (USA), placed under the tongue for 1 minute, at 8 designated time points (Figure 1). Oral swabs were selected due to ease of data collection in the athletic field and their noninvasiveness when compared with venepuncture.[11] In the researcher's absence, participants collected the

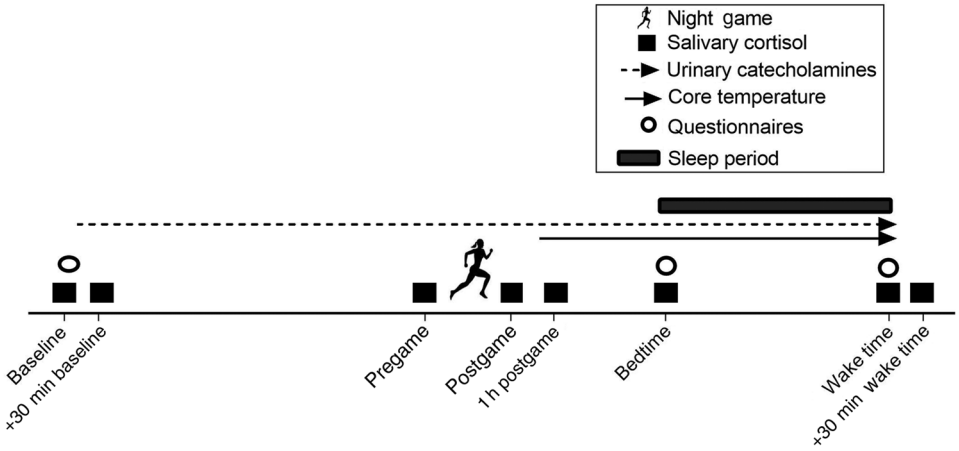

Figure 1 — An illustration of the collection time points for each variable measured. Game measures were time matched on the rest day.

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overnight samples at each designated time point and documented the collection time to ensure adherence to time in the subsequent trial. Once received, saliva samples were stored at −20°C as instructed by the manufacturer (Stratech Scientific APAC Pty Ltd, Sydney, Australia) and analyzed within 2 months of collection. Salivary cortisol was measured using a commercially available ELISA assay (Salimetrics, Carlsbad, CA). The intraassay variance was 3.1%, and the sensitivity of the assay was 0.003 µg/dL.

Twenty-Four-Hour Urinary Catecholamine. Catecholamine excretion was measured from urine samples collected during 2 time periods over 24 hours, adapted from Netzer et al.[17] The first time period was from awakening after emptying bladder to the start of the netball game. The second was immediately postnetball game until awakening the next day. The time points recorded from the night game were time matched on the rest day. A commercial lab analyzed the urine collection for adrenaline and noradrenaline using high-performance liquid chromatography with electrochemical detection. Catecholamines were measured as an excretion rate and reported in nmol/L. Urinary noradrenaline levels are considered a reliable indicator of overall sympathetic activity, whereas adrenaline levels reflect the adrenal response.[17]

Psychometric Assessment

A modified version of the Karolinska Sleep Questionnaire was used to subjectively assess sleep quality from the previous night's sleep as described by Garde et al[12] and Åkerstedt et al.[18] In total, 7 items—(1) difficulties falling asleep, (2) disturbed/restless sleep, (3) repeated awakenings, (4) premature awakenings, (5) difficulties waking up, (6) nonrefreshing sleep, and (7) exhausted at awakening—were scored between 1 (0 awakenings) and 5 (4 or more awakenings). A disturbed sleep index (Q1–4) and an awakening index (Q5–7) were calculated, with high scores representing a more disturbed sleep and less sleep satisfaction upon awakening.[12] Responses were calculated for the morning of the collection day and the morning following.

The PSAS was used to measure subjective presleep arousal at bedtime in participants the night prior to testing and the night following testing.[19] The PSAS is a 16-item self-reported scale that assesses somatic ("a jittery, nervous feeling in your body") and cognitive ("worry about falling asleep") manifestations of arousal.[19] Participants rated how intensely they experienced each item as they tried to fall asleep from the previous night on a scale from 1 = "not at all" to 5 = "extremely." All answers were calculated and divided into a somatic and cognitive domain, with a minimum score of 8 and a maximum possible score of 40.

To assess trait arousal, participants completed the hyperarousal scale 1 week prior to the night game (when the actigraph was distributed). The scale is commonly utilized in patients with insomnia due to their increased cortical responsiveness that is less modulated by inhibitory processing.[20] The hyperarousal scale contained 26 items for which participants selected 1 of 4 responses (0 = not at all, 1 = a little, 2 = quite a bit, and 3 = extremely). Completed item scores were aggregated to yield an overall hyperarousal score. Submeasures included an extreme score (number of items checked "extreme"), an introspectiveness score (summed score for items "My mind is always going," "I think a lot about feelings," "I tend to anticipate problems," "I take things personally," "Some thoughts return too often," and "I take a long time to make decisions"), and a react score (sum score for items "Bright lights, crowds, noises of traffic bother me," "I get rattled when a lot happens at once," and "A sudden loud noise would cause me a prolonged reaction").[20]

Statistical Analysis

Data are expressed descriptively as means (SD) of the individual data and as single individual data. Statistical evaluation was completed using paired sample *t* tests for significance of differences between the night-game and rest-day sleep data. Differences in cortisol, core temperature, and catecholamines between condition (night game and rest day) and time of day were analyzed using a 2-way analysis of variance with repeated measures. Significant main effects or interactions were further analyzed using Fisher's least significant difference post hoc analyses. Standardized effect sizes (Cohen's *d*) were calculated to interpret the magnitude of the mean differences between data with $d < 0.20$ (trivial), $d = 0.20$ (small), $d = 0.50$ (medium), and $d > 0.80$ (large). A Wilcoxon signed-rank test analyzed arousal and subjective sleep measures to test for differences between night game and rest day. To satisfy the assumption of correlation analysis, a Spearman's correlation coefficient (r_x) was utilized to explore associations between physiological, neuroendocrine, and psychometric variables and sleep data for the night-game and rest-day data.[17] All statistics were completed using SPSS© Statistics software program (version 19; IBM©, New York, NY) with significance set to $P \leq .05$.

Results

Objective Sleep Variables

Mean and individual data for sleep efficiency and sleep duration following the night game and rest day are illustrated in Figure 2. Sleep duration for the night-game condition was less compared with the rest day ($P = .02$, $d = 1.13$). Participants experienced reduced sleep efficiencies following the night game when compared with a rest day ($P < .001$, $d = 0.73$). No differences were evident between conditions for bedtime, sleep latency, and number of wake bouts in the sleep period measured ($P > .05$; Table 1). Participants woke

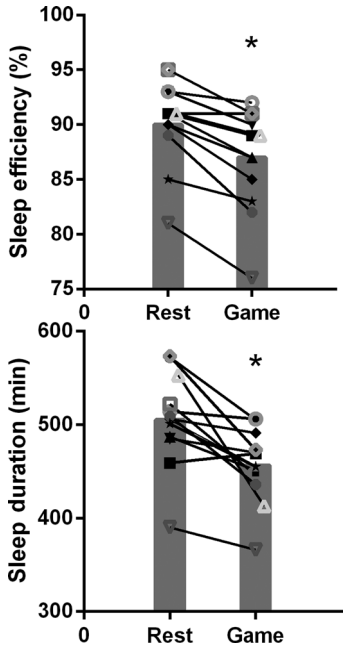

Figure 2 — Mean (shaded bars) and individual participant's (n = 12; symbols) sleep efficiency and sleep duration for the night following a rest day and the night game. *Significant differences between the night game and rest day ($P < .05$).

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Table 1 Mean (SD) and Effect Sizes (Cohen *d*) for Sleep, Physiological, and Psychological Measures Comparing a Time-Matched Rest Day to a Night Game in Netballers

	Condition		
	Rest day	Night game	Cohen *d*
Sleep variables			
Bedtime, h:min	22:32 (00:55)	22:27 (00:52)	0.09
Wake time morning after testing, h:min	07:55 (00:34)	07:08 (00:34)*	1.38
Sleep latency, min	14.5 (15.1)	24.4 (27.4)	0.45
Number of wake bouts	19 (7)	19 (7)	0.00
Core temperature, °C			
Postexercise	37.42 (0.18)	38.81 (0.51)*	3.63
Bedtime	37.24 (0.31)	37.18 (0.37)	0.18
Waking	37.10 (0.27)	37.00 (0.31)	0.34
Noradrenaline, nmol/L**			
Pregame	97.83 (36.43)	232.6 (148.1)	1.25
Postgame	88.58 (54.08)	234.1 (137.2)	1.40
Adrenaline, nmol/L**			
Pregame	12.91 (5.71)	26.92 (15.88)	1.17
Postgame	11.92 (4.56)	18.67 (13.26)	0.68
Psychometric variables			
Presleep arousal—somatic			
Night before game	10.0 (1.8)	10.8 (3.8)	0.27
Night after game	9.4 (1.4)	10.3 (4.3)	0.28
Presleep arousal—cognitive			
Night before game	14.3 (3.1)	16.8 (7.0)	0.46
Night after game	13.4 (3.9)	14.6 (6.2)	0.23
Disturbed sleep index			
Morning of the game	8.1 (2.8)	8.7 (2.8)	0.21
Morning postgame	6.3 (2.6)	7.4 (3.4)	0.36
Awakening index			
Morning of the game	5.6 (2.2)	7.1 (2.7)	0.61
Morning postgame	4.4 (2.0)	5.3 (2.9)	0.36

Note: All night before-game and after-game time points were time matched on the rest day. $d < 0.2$, trivial; $d = 0.2$, small; $d = 0.5$, medium; $d > 0.8$, large.
*Significant difference between night-game and rest-day conditions ($P < .05$). **Significant main effect for condition; the night game greater than the rest day ($P < .05$).

earlier following the night game when compared with a rest day ($P \leq .01$, $d = 0.959$). No differences were found between sleep metrics for the rest day and 2-week sleep data ($P > .05$).

Psychometric Variables

From the sleep diary, 8 participants (67%) experienced worse subjective sleep (average) following the night game compared with a rest day (good), whereas 4 participants (33%) experienced no change. There was an increase in the median subjective sleep rating from the sleep diary to reflect poor sleep ($P \leq .01$, $d = 1.03$). No differences ($P > .05$) were found for calculated variables in the Karolinska Sleep Questionnaire and the PSAS between conditions (Table 1).

Physiological and Neuroendocrine Variables

A main effect for condition ($P < .01$) was found for noradrenaline levels, with participants experiencing elevated levels of noradrenaline in the night-game condition when compared with the rest day (Table 1). Adrenaline levels also demonstrated a main effect for condition ($P = .02$), with levels for the night-game condition elevated compared with the time-matched rest day. An interaction was observed for measures of cortisol, with greater baseline levels measured in the morning of the night game when compared with a time-matched rest day ($P = .027$, $d = 0.60$; Figure 3). In addition, greater cortisol measures were observed immediately postgame and 1 hour postgame when compared with the time-matched rest day ($P = .006$, $d = 1.49$ and $P = .031$, $d = 0.25$, respectively; Figure 3). No differences were observed for the cortisol awakening response the morning of the game and the morning following the game compared with a rest day. An interaction was observed for core temperature, with greater temperatures measured immediately postgame in the night-game condition compared with the rest day ($P < .01$, $d = 3.63$). No differences were found for bedtime and awakening core temperatures before and after the night game compared with a rest day (Table 1).

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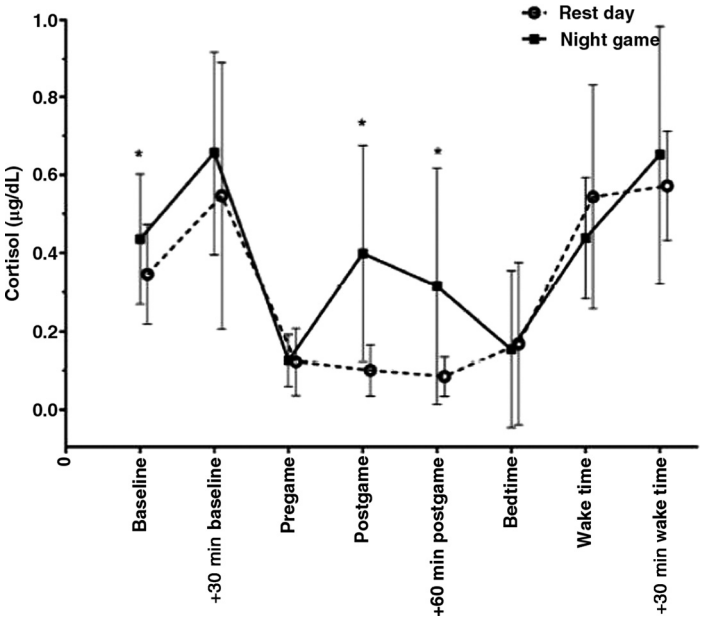

Figure 3 — Mean salivary cortisol values collected at 8 designated time points a day before and after a night game and a time-matched rest day in netballers. *Significant difference found at the time point between conditions (rest day and the night game; $P < .05$).

Correlation Between Sleep and Physiological, Neuroendocrine, and Psychometric Variables

Significant correlations were found for trait psychometric measures between the hyperarousal scale and sleep efficiency. A decrease in sleep efficiency was strongly associated with an increase in hyperarousal scores (trait) in netballers, $r_s(23) = -.611$, $P \leq .01$. A moderate negative correlation was identified between sleep efficiency and both the extreme score and introspectiveness score calculated from the hyperarousal scale, $r_s(23) = -.541$, $P \leq .01$ and $r_s(23) = -.476$, $P > .05$, respectively. No correlations were found between any physiological and neuroendocrine variables and sleep (efficiency, duration, and sleep onset latency).

Discussion

This study examined the impact of nighttime competition on sleep in elite netballers. The main study findings were as follows: (1) participants experienced reduced sleep durations, lower sleep efficiency, early awakenings, and poorer subjective sleep ratings following the night game compared with the rest day; (2) neuroendocrine urinary catecholamines (adrenaline and noradrenaline) were elevated before and after the night game in comparison with a time-matched rest day; (3) salivary cortisol concentrations were elevated at the morning of the game, immediately postgame, and 1 hour postgame in comparison with a rest day; and (4) correlation analysis revealed a strong negative correlation between sleep efficiency and the hyperarousal trait score.

Following nighttime competition, a decrease in both objective and subjective measures of sleep occurred (Table 1; Figure 2), which is consistent with recent findings in team sport athletes.[2–4] When compared with a rest day, 92% (11 out of 12) of participants experienced reduced sleep following the night game, with objectively measured (eg, actigraphy) sleep durations decreasing in the range of 15 to 140 minutes (Figure 2). These findings are comparable to reported reductions of sleep quantity in male footballers (181 min)[4] and Australian rules footballers (171 min and 126 min) following night games.[5,6] The observed decrease in sleep duration following the night game was not a consequence of delayed bedtime, but instead due to earlier wake times, which were ad libitum. Our findings are not completely consistent with Fullagar et al[4] who, in addition to earlier wake times, observed later bedtimes in male football players following a night match compared with a normal training day. We suspect that the differences between these studies were likely due to logistical considerations enabling our athletes to avoid delaying bedtime. For instance, participants in the current study did not have media or recovery commitments after the game and were close to their residential facilities from the sport complex (approximately 800 m).

Regardless, the current results suggest that, under normal circumstances (rest day), the netballers obtained adequate sleep durations (Figure 2), which are consistent with recommendations for healthy adults,[21] and irrespective of condition, sleep efficiency was greater than the clinically relevant 85%, indicative of poor sleep efficiency.[22] Despite this, following late competition, sleep needs were not met even though participants specified their need to obtain 9 hours of sleep a night to feel fully rested. This finding is particularly important for netballers, as it is common in major tournaments for athletes to play 8 games within 10 days; therefore, the ability to maximize sleep as a form of recovery is vital.

Compared with the rest day, noradrenaline and adrenaline levels were significantly elevated for the night-game condition, as was cortisol concentration the morning of the game (baseline), immediately following the game, and 1 hour postgame when time matched to the rest day (Figure 3). Similar findings have been observed for cortisol levels in 16 tennis players, where a competition day induced a higher level of stress when compared with a rest day.[23] Furthermore, Pierce et al[24] reported elevated epinephrine levels in track and field athletes vying for international selection compared with team members who were not, highlighting the anticipatory stress effect competition can impose on an athlete. The current study was a selection and an international game for participants; therefore, elevated levels of stress hormones prior to the game may indicate an anticipatory affect, whereas continued elevation after the game may likely be due to sustained psychological stress in addition to stress from physical activity.[24] The notion that the netball competition caused a degree of physiological stress is therefore supported through the elevation of peripheral stress hormones[23,25]; however, a lack of correlation between sleep measures and any stress hormone indicates the absence of association.

Despite increased cortisol following the night game, levels measured immediately before bedtime (~3 h and 15 min postgame) were similar between conditions. Our findings are consistent with Elloumi et al,[25] who observed a 2.5-fold increase in cortisol levels in 20 rugby players immediately following a match (4:00 PM); yet, reported cortisol levels had returned to basal values within 4 hours (8:00 PM). Decreased cortisol levels from postgame to bedtime, although speculative, may perhaps represent the flexibility of an athlete's autonomic nervous system to return to basal levels within hours following a competition. Furthermore, in contrast to patients with insomnia, where elevated cortisol levels are consistent with poor sleep,[26] our findings indicate cortisol is not likely to influence the sleep of athletes following night games when the time frame between the end of competition and sleep onset is greater than 3 hours. A similar finding was observed for core temperature, a mechanism hypothesized to influence sleep following night

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competition.[6] Core temperature was elevated immediately postgame compared with the time-matched day (Table 1), yet no differences were evident between conditions immediately prior to bed. Therefore, like cortisol, our findings provide evidence against the proposed assumption linking elevated core temperature and sleep following night competition.[6]

Psychometric state measures indicated no differences between conditions. This finding is in contrast to a similar study in healthy working populations that found psychological arousal at bedtime to be positively associated with reports of poor sleep in healthy workers.[12] Although the current study results indicate athletes may not have impaired sleep due to psychometric state arousal (PSAS; Table 1), correlation analysis indicates a high trait arousal (hyperarousal scale) is strongly associated with a decrease in sleep efficiency. Thus, netballers who rate higher in the hyperarousal scale may potentially be predisposed to disturbed sleep around critical competition times or under certain stressful situations. Previous experimental studies in patients with sleep disorders and civil servants have suggested that the underlying mechanisms behind sleep problems may be due to hyperarousal.[27] Indeed, the hyperarousal scale has previously been shown to correlate highly to several electroencephological variables, signifying a tendency of individuals to exhibit cortical electrical responses that are less modulated by stimulus properties. This reduced ability to modulate consequently results in an inability to cortically de-arouse therefore serving as a vulnerability to poor sleep.[20,28] Although the suggestions are based on inferences from other populations, the similarity of the results to the current athlete group could provide insight into an area that has not previously been explored.

Conclusions

This study's findings confirm athletes have a reduced duration and quality of sleep following a night game. To the authors' knowledge, this is the first study to explore the hypothesized mechanisms proposed for poor sleep in athletes following late competitions. Results suggest neuroendocrine and physiological variables are not related to poor sleep following a night game when sleep is more than 3 hours postcompetition. However, correlation analysis suggested athletes who had a tendency toward a high trait arousal (hyperarousal scale) may be susceptible to sleep complaints following a night game. Given that the occurrence of night games is common in sport, our findings provide information that is useful for athletes, coaches, and support staff when devising individualized recovery plans.

Limitations

The limitations of this study should be considered when interpreting results. First, the sample size employed in the current study was a sample of convenience due to the applied setting, which may have limited the significance of the results; however, this is not uncommon in studies with elite-level athletes. Second, numerous factors are known to influence the sleep characteristics of athletes. Although many factors were measured in this study, it may be beneficial for future research to explore the match day load interaction with sleep characteristics.

Practical Applications

Our results demonstrate that athletes experience worse sleep following night competitions but obtain adequate sleep under normal conditions (rest day). A novel contribution is that trait psychometric measures appear to predispose athletes to poor sleep following a night competition and refute commonly hypothesized explanations (neuroendocrine variables and core temperature). Practitioners should be aware of this occurrence and look to alleviate poor sleep through targeted interventions addressing hyperarousal in susceptible athletes.

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